

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION**Revision Test 2026****Chemistry — Class 9 (Science Group)**

Syllabus: Chapter 1 – Nature of Chemistry in Science | Chapter 2 – Matter | Chapter 3 – Atomic Structure | Chapter 4 – Periodic Table & Periodicity of Properties | Chapter 5 – Chemical Bonding | Chapter 6 – Stoichiometry | Chapter 7 – Electrochemistry | Chapter 8 – Energetics | Chapter 9 – Chemical Equilibrium

Total Marks: 50**Time: 2 Hours 15 Minutes****Paper: Exam-22 (Version 1)**

1. Attempt **ALL** questions. There are no choices in this paper.
2. Section A (10 marks): 10 MCQs. Fill the correct bubble. Suggested time: 15 minutes.
3. Show complete working for every calculation. Always write the correct unit with your final answer.
4. Read every part of a question and answer **all** of it before moving on. Pencil is allowed only for diagrams.

SECTION A — Multiple Choice Questions (10 × 1 = 10 Marks)

#	Q.1 Choose the best possible option.	A	B	C	D	Answer
1	The process of applying scientific knowledge to practical applications, resulting in the creation of tools, machines and systems, is called:	science	technology	engineering	chemistry	(A)(B)(C)(D)
2	In diamond, each carbon atom is covalently bonded to:	three other carbon atoms in flat hexagonal layers	two other carbon atoms in a long chain	sixty other carbon atoms in a hollow ball	four other carbon atoms in a rigid tetrahedral network	(A)(B)(C)(D)
3	Which statement belongs to Bohr's model of the atom but NOT to the quantum mechanical model?	Electrons revolve around the nucleus in fixed circular orbits of definite energy.	Electrons are found in probability regions called orbitals.	Electrons behave as both particles and waves.	The exact path of an electron cannot be mapped.	(A)(B)(C)(D)
4	When a nucleus emits a NEUTRON during radioactive decay, the atom changes into:	another element	a positive ion of the same element	another isotope of the same element	a negative ion of the same element	(A)(B)(C)(D)
5	The amount of energy RELEASED when an electron is added to the valence shell of an isolated gaseous atom, forming a uni-negative ion, is called:	electron affinity	ionization energy	electronegativity	shielding effect	(A)(B)(C)(D)
6	The transition metal used as a catalyst in the manufacture of margarine is:	iron	nickel	platinum	palladium	(A)(B)(C)(D)
7	Silicon dioxide (SiO ₂) is very hard and difficult to melt because it has:	small separate molecules held together by weak intermolecular forces	a sea of delocalized electrons around positive ions	oppositely charged ions held in a lattice	a giant covalent structure in which each atom is joined to several others by	(A)(B)(C)(D)

					strong covalent bonds	
8	In the formula $(\text{NH}_4)_2\text{SO}_4$, the subscript 2 written outside the bracket shows that one formula unit contains:	two sulfate ions	two ammonium ions	one ammonium ion carrying a charge of +2	two hydrogen atoms	(A)(B)(C)(D)
9	The removal of hydrogen from a substance is called:	reduction	corrosion	oxidation	electroplating	(A)(B)(C)(D)
10	When a few drops of water are added to blue anhydrous cobalt(II) chloride, it turns:	pink	white	colourless	yellow	(A)(B)(C)(D)

SECTION B — Short Questions (6 × 4 = 24 Marks)

Q.2	Attempt all six parts.	Marks									
(i)	The table shows the solubility of two solutes in grams per 100 g of water (data from your textbook). <table border="1"> <thead> <tr> <th>Solute</th><th>20 °C</th><th>100 °C</th></tr> </thead> <tbody> <tr> <td>NH_4Cl</td><td>37.5 g</td><td>77 g</td></tr> <tr> <td>$\text{Ca}(\text{OH})_2$</td><td>0.173 g</td><td>0.066 g</td></tr> </tbody> </table> (a) Define a saturated solution. [1] (b) Define an unsaturated solution. [1] (c) Using the data, describe how an increase in temperature affects the solubility of each solute. [2]	Solute	20 °C	100 °C	NH_4Cl	37.5 g	77 g	$\text{Ca}(\text{OH})_2$	0.173 g	0.066 g	4
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(ii)	Bromine occurs naturally as two isotopes: ^{79}Br with an abundance of 50.5% and ^{81}Br with an abundance of 49.5%. (a) Calculate the relative atomic mass of bromine, showing your working. [2] (b) State the number of protons, neutrons and electrons in one atom of ^{81}Br (atomic number 35). [1] (c) Explain why the two isotopes of bromine have the same chemical properties. [1]	4									
(iii)	Two elements have the following electronic configurations: X: $1s^2 2s^2 2p^6 3s^2$ Y: $1s^2 2s^2 2p^6 3s^2 3p^4$ (a) State the period and the group of X and of Y, giving a reason for each. [2] (b) Write the charge on the ion formed by X and by Y. [1] (c) Using the symbols X and Y, write the formula of the compound they form. [1]	4									
(iv)	Hydrogen chloride, HCl, is a covalent compound. (Atomic numbers: H = 1, Cl = 17) (a) Draw a dot-and-cross diagram of one molecule of HCl, showing all the outer-shell electrons of chlorine. [2] (b) State whether the H–Cl bond is polar or non-polar, and explain why. [2]	4									
(v)	Consider the reaction: $2\text{NH}_3 + 3\text{CuO} \rightarrow 3\text{Cu} + \text{N}_2 + 3\text{H}_2\text{O}$ (a) Find the oxidation number of copper in CuO and in Cu. [1] (b) Find the oxidation number of nitrogen in NH_3 and in N_2. [1] (c) Which element is oxidized and which is reduced? Give a reason based on oxidation numbers. [1] (d) Name the oxidizing agent and the reducing agent in this reaction. [1]	4									
(vi)	Sulphur dioxide and oxygen are mixed in a sealed container: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ (a) Which is the forward reaction and which is the reverse reaction? [1] (b) Why is the rate of the reverse reaction zero at the very start? [1] (c) State two things that are true once the system has reached equilibrium. [2]	4									

SECTION C — Long Questions (2 × 8 = 16 Marks)

Q.3	Attempt both parts.	Marks
(i)	Nitrogen reacts with oxygen at very high temperature: $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}(\text{g})$ Bond energies: $\text{N}\equiv\text{N} = 945 \text{ kJ/mol}$, $\text{O}=\text{O} = 498 \text{ kJ/mol}$, $\text{N}=\text{O} = 630 \text{ kJ/mol}$. (a) Calculate the enthalpy change of this reaction from the bond energies. Show the energy for bonds broken and bonds formed. [3] (b) State whether the reaction is exothermic or endothermic, and give a reason. [1] (c) Draw a labelled reaction pathway (energy) diagram for this reaction. Label the reactants, the products, the enthalpy change, and the activation energy both without and with a catalyst. [3] (d) State one thing a catalyst changes in this reaction and one thing it does not change. [1]	8
(ii)	A 2.1 g sample of a gaseous hydrocarbon P contains 1.8 g of carbon and 0.3 g of hydrogen. Its relative molecular mass is 42. (C = 12, H = 1, O = 16) (a) Calculate the number of moles of carbon and of hydrogen in the sample, and hence find the empirical formula of P. [2] (b) Find the molecular formula of P. [1] (c) Calculate the number of moles in 8.4 g of P. [1] (d) Calculate the number of molecules in 8.4 g of P. [1] (e) P burns completely in oxygen to form carbon dioxide gas and water. Write the balanced equation for this reaction, with state symbols, given that water is formed as a liquid. [2] (f) Calculate the mass of one molecule of P in grams. [1]	8

Chapters covered: Chemistry Class 9 — Chapters 1 to 9.

Candidate's Signature: _____

Invigilator's Signature: _____

— End of Question Paper —